Project Title: **Door Buzzer Using Ultrasonic Sensor**

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Domain: Internet of Things (IoT)
Technologies Used: Arduino Uno, Ultrasonic Sensor, Buzzer
Project Type: Academic / Mini Project
Year: 2024

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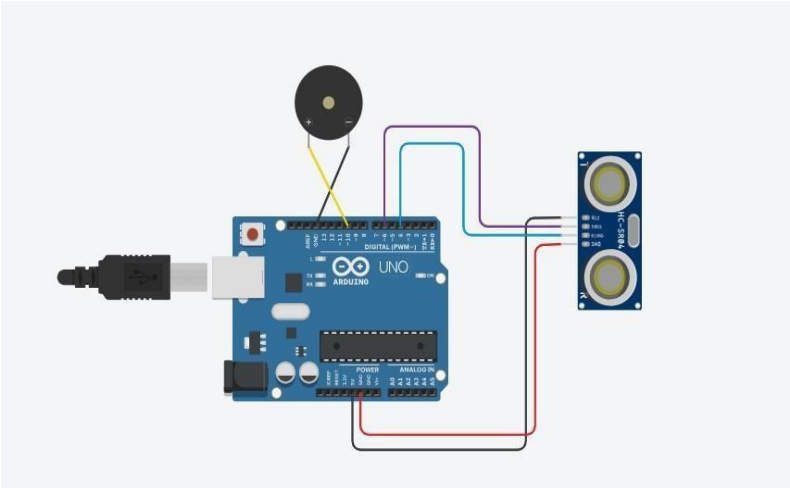
AIM:

The aim of this project is to design and implement a door buzzer system utilizing an ultrasonic sensor. The primary objective is to create a device that detects the presence of individuals approaching the door and triggers a buzzer alarm to alert occupants of their presence. The system will be designed to accurately measure the distance of approaching objects, differentiate between authorized and unauthorized individuals based on predefined distance thresholds, and activate the buzzer accordingly. Additionally, the project aims to explore the practical applications of ultrasonic sensors in enhancing security and convenience in residential or commercial settings.

ABSTRACT:

Covid-19 has created havoc in people also we can see it is increasing day by day all over the world. We also know a well known proverb that “prevention is better than cure” so, we need to search for prevention methods so that it can be avoided. Therefore, some of the preventive measures are as follows: social distancing, avoiding surface contact, washing hands frequently and wearing mask. Then also there are many unavoidable circumstances like people visiting to the homes, in that case the guests have to ring the doorbell which might also spread the infection. Thus, as the doorbell has also become hotspot for spread of the virus this can be avoided if we use touch less doorbell. By which, we can avoid surface contact. Touch less door Bell can be created using an ultrasonic sensor and Arduino Uno. In which, we just have to put our hand in front of the sensor then the buzzer rings to intimate the owner of the house.

CIRCUIT DIAGRAM:



COMPONENTS REQUIRED:

Name	Quantity	Component
U1	1	Arduino Uno R3
PIEZO1	1	Piezo
DIST1	1	Ultrasonic Distance Sensor (4 – pin)

- **Arduino Uno R3:**

The Arduino Uno R3 is a versatile microcontroller board popular for DIY projects. It features 14 digital I/O pins, 6 analog inputs, USB programming interface, and compatibility with a wide range of sensors and shields.

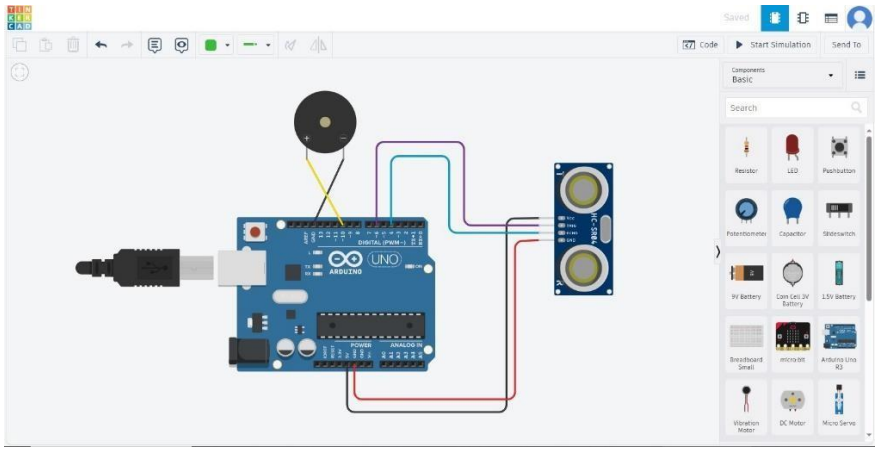
- **Piezo:**

Piezoelectric transducers convert mechanical energy into electrical signals and vice versa. They are used in buzzers, sensors, actuators, and energy harvesting devices.

- **Ultrasonic Distance Sensor (4 – pin):**

The 4-pin ultrasonic distance sensor measures distance using ultrasonic waves. It emits a pulse, measures the time it takes for the pulse to bounce back, and calculates distance. It's commonly used in robotics and automation for obstacle detection and navigation.

TINKERCAD DIAGRAM:

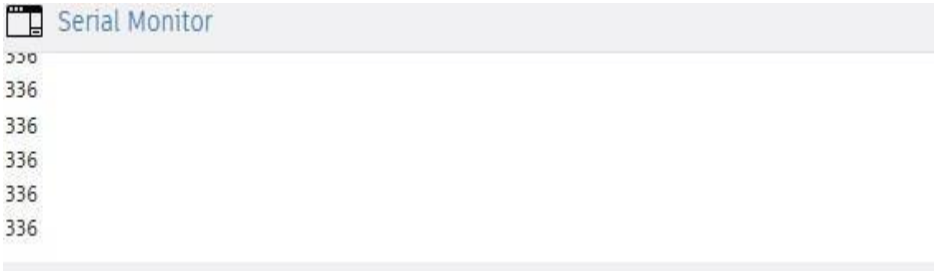


PROGRAM:

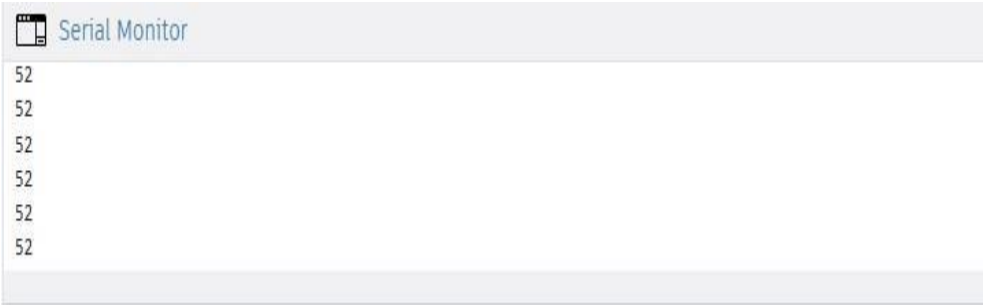
```cpp
// C++ code
//
int ultrasonic = 0;

long readUltrasonicDistance(int triggerPin, int echoPin)
{
  pinMode(triggerPin, OUTPUT);  // Clear the trigger
  digitalWrite(triggerPin, LOW);
  delayMicroseconds(2);
  // Sets the trigger pin to HIGH state for 10 microseconds
  digitalWrite(triggerPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(triggerPin, LOW);
  pinMode(echoPin, INPUT);
  // Reads the echo pin, and returns the sound wave travel time in microseconds
  return pulseIn(echoPin, HIGH);
}

void setup()
{
  Serial.begin(9600);
  pinMode(10, OUTPUT);
  pinMode(10, OUTPUT);
}

void loop()
{
  ultrasonic = 0.01723 * readUltrasonicDistance(6, 4);
  Serial.println(ultrasonic);
  if (ultrasonic < 115) {
    tone(10, 523, 100); // play tone 60 (C5 = 523 Hz)
  } else {
    digitalWrite(10, LOW);
  }
  delay(10); // Delay a little bit to improve simulation performance
}
```

SERIAL MONITOR (OUTPUT):

When the distance of obstacle increase and also the value will increase.



And when the distance of obstacle decrease and also the value will decrease.



RESULT:

- Distance measurements were successfully obtained using the ultrasonic sensor.
- The buzzer reliably activated when an object approached within the specified range (e.g., 1 meter).
- During testing, the system demonstrated consistent performance in detecting movement and triggering the buzzer.